

## NuMicro® MS51 Series

Small form factor microcontroller  
for smart home appliances



NuMicro® MS51 series microcontrollers based on enhanced 1T 8051 core offer a range of memory with 8/ 16/ 32 Kbytes Flash and 1/ 2 Kbytes SRAM, providing multiple package types from 10 to 33 pins, 2.5V to 5.5V operating voltage, and -40°C to 105°C industrial temperature range. In MS51, there are 10 kHz and 24 MHz internal clock generators and each I/O pin is able to provide 20 mA high sink current.

- Enhanced 1T 8051 core up to 24 MHz
- Supports 8/ 16/ 32 Kbytes Flash and 1/ 2 Kbytes SRAM
- 128 Bytes Secure Protection ROM (SPROM)
- Up to 15 channels 12-bit ADC to measure  $V_{dd}$  by bandgap voltage reference
- Up to 12 channels standalone PWM with dead zone insertion and complimentary output
- Supports high noise immunity: 8 kV ESD and 4.4 kV EFT to ensure high reliability in harsh conditions
- Supports small factor packages: MSOP10, TSSOP14, TSSOP20, QFN20, TSSOP28, and QFN33